Striping & layout settings — every knob

Setting	What it does	Range	Sane default	The trade
`stripe_count`	how many OSTs a file spreads over	1 → # of OSTs (-1 = all; max 2000)	**1** (the trap)	bandwidth ↑ vs blast-radius ↑
`stripe_size`	bytes written to one OST before moving on	64 KiB → 4 GiB	1–4 MiB	small = more overhead; large = less parallelism on small files
`layout / PFL`	stripe pattern changes with file size	single → composite (progressive)	single, until it hurts	policy follows file size, no user guesswork
`DoM size`	tiny-file tail kept on MDT flash	multiple of 64 KiB	off, until small-file heavy	small-file latency ↓ vs scarce MDT flash
`pool`	restrict a file to a named subset of OSTs	any defined pool	(none)	media control (flash vs capacity) per path

Every knob is a trade: bandwidth vs. overhead vs. resilience. Ranges per EXAScaler 6.3.6 Admin Guide / Understanding Lustre. Inspect with `lfs getstripe`*; set with* `lfs setstripe`*.*